VARDHAMAN
COLLEGE OF ENGINEERING

A Course End Project Report on

Employee Salary Data Analysis

A8518- Python For Data Analysis

Submitted by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

This is to certify that the Course End Project titled **"Employee Salary Data Analysis"** is carried out by **C. AKHILA** with Roll Number **24881A05E4** towards **A8518 – Python For Data Analysis Laboratory** course in partial fulfilment of the requirements for the award of degree of **Bachelor of Technology** in **Computer Science and Engineering** during the Academic year 2025-26.

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Abstract

Employee Salary Data Analysis is a comprehensive data-driven approach aimed at understanding patterns, trends, and relationships in employee compensation within an organization. This project focuses on analyzing key attributes such as age, experience, department, job role, and salary to determine how these factors influence pay distribution. By examining these variables, the analysis helps identify salary trends, disparities, and potential inequalities among employees. Such insights are crucial for organizations to ensure fair compensation practices, maintain transparency, and enhance employee satisfaction and retention.

The project follows a structured methodology that includes data collection, preprocessing, exploratory data analysis (EDA), and visualization. Data preprocessing ensures accuracy by handling missing values and inconsistencies, while EDA techniques help uncover hidden patterns and relationships. Visualization tools such as histograms and bar charts are used to represent salary distributions and departmental comparisons clearly. The findings reveal that experience and job roles significantly impact salary levels, while variations across departments highlight the need for balanced pay structures. Overall, this project demonstrates how data analytics can support informed decision-making in human resource management and contribute to building an efficient and equitable workplace.

Keywords: Employee Salary Analysis, HR Analytics, Data Analytics, Exploratory Data Analysis (EDA), Salary Distribution, Data Visualization, Pandas, NumPy, Matplotlib, Workforce Management

1 Introduction

In today's data-driven world, organizations rely heavily on data analytics to make informed decisions, especially in the field of human resource management. One of the most critical aspects of HR management is employee compensation, as it directly impacts employee satisfaction, motivation, and retention. Employee Salary Data Analysis involves examining salary-related data along with various employee attributes such as age, experience, department, job role, and performance to understand patterns and trends in compensation. By analyzing this data, organizations can ensure fair pay structures, identify salary disparities, and improve overall workforce management strategies.

With the increasing availability of employee data, companies have the opportunity to use analytical techniques to gain deeper insights into salary distribution. Different factors influence employee salaries; for instance, employees with more experience or higher qualifications tend to receive higher compensation. Similarly, salary variations may exist across departments depending on job roles and responsibilities. Without proper analysis, such variations may lead to dissatisfaction, reduced productivity, and increased employee turnover. Therefore, it becomes essential for organizations to study salary data systematically to maintain transparency and fairness in compensation practices.

Employee Salary Data Analysis also plays a vital role in identifying inequalities within an organization, such as pay gaps based on gender, department, or experience level. Detecting and addressing these issues helps organizations build a more inclusive and equitable work environment. Additionally, understanding salary trends enables companies to design better compensation policies, offer competitive salaries, and attract skilled talent. It also supports decision-making in areas such as promotions, bonuses, and performance evaluations.

This project focuses on analyzing employee salary data using data analytics techniques and visualization tools. The process begins with data collection, followed by preprocessing steps such as handling missing values and organizing the data into a structured format. Exploratory Data Analysis (EDA) is then performed to identify patterns, relationships, and trends in the dataset. Visualization techniques such as histograms, bar charts, and correlation analysis are used to present the findings in a clear and meaningful way.

The implementation of this project utilizes Python and its powerful libraries, including Pandas for data manipulation, NumPy for numerical operations, and Matplotlib for visualization. These tools

enable efficient handling of data and help in generating accurate insights. By applying these techniques, the project demonstrates how employee salary data can be analyzed effectively to support data-driven decision-making. Overall, this study highlights the importance of using analytics in human resource management to improve organizational efficiency, ensure fair compensation, and enhance employee satisfaction.

2 Literature Review

Employee Salary Data Analysis has become an important area of research with the growing adoption of data analytics in human resource management. Several studies highlight that employee compensation is influenced by multiple factors such as experience, education, job role, and organizational structure. Researchers have emphasized that analyzing salary data helps organizations understand compensation patterns and design fair pay structures. It has been observed that employees with higher experience and specialized skills tend to receive better salaries, while variations across departments are often based on the nature of work and responsibilities. These insights encourage organizations to adopt data-driven approaches for managing employee compensation effectively.

Many researchers have focused on the use of Exploratory Data Analysis (EDA) and statistical techniques to examine salary datasets. EDA is widely used to understand the distribution of salaries, detect outliers, and identify relationships between variables such as experience and salary. Techniques like correlation analysis, grouping, and aggregation help in uncovering meaningful patterns in employee data. In addition, machine learning models have also been applied in some studies to predict employee salaries based on input features like qualifications and years of experience. These approaches not only improve accuracy but also assist organizations in forecasting salary trends and making informed decisions.

Furthermore, data visualization plays a significant role in salary analysis by simplifying complex data into easily interpretable formats. Graphical representations such as histograms, bar charts, and heatmaps help in identifying salary distributions and differences across various employee groups. Researchers also stress the importance of data preprocessing techniques, including handling missing values, removing duplicates, and ensuring consistency, to improve the reliability of analysis. Tools like Python, along with libraries such as Pandas, NumPy, and Matplotlib, are widely used for performing efficient data analysis and visualization. Overall, existing studies demonstrate that salary data analysis is essential for ensuring fairness, improving employee satisfaction, and supporting strategic decision-making in organizations.

3 Objectives

- **To Analyze Employee Salary Distribution**

The primary objective of this project is to examine how salaries are distributed among employees based on factors such as age, experience, and department. By analyzing salary data, the project aims to identify patterns, trends, and variations in compensation. This helps in understanding how different employee attributes influence salary levels and whether the distribution is balanced across the organization.

- **To Identify Relationships Between Employee Attributes and Salary**

Another important objective is to study the relationship between salary and various employee characteristics such as experience, job role, and department. Through techniques like correlation analysis and grouping, the project aims to determine how strongly these factors impact salary. This helps in identifying key factors that contribute to higher or lower compensation within the organization.

- **To Derive Insights for Better Decision-Making**

The final objective is to generate meaningful insights that can support organizational decision-making. By using data analysis and visualization techniques, the project aims to provide clear conclusions regarding salary trends and disparities. These insights can help organizations design fair salary structures, improve employee satisfaction, and make data-driven decisions related to promotions, bonuses, and workforce planning.

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4 Methodology

4.1 Architectural Diagram

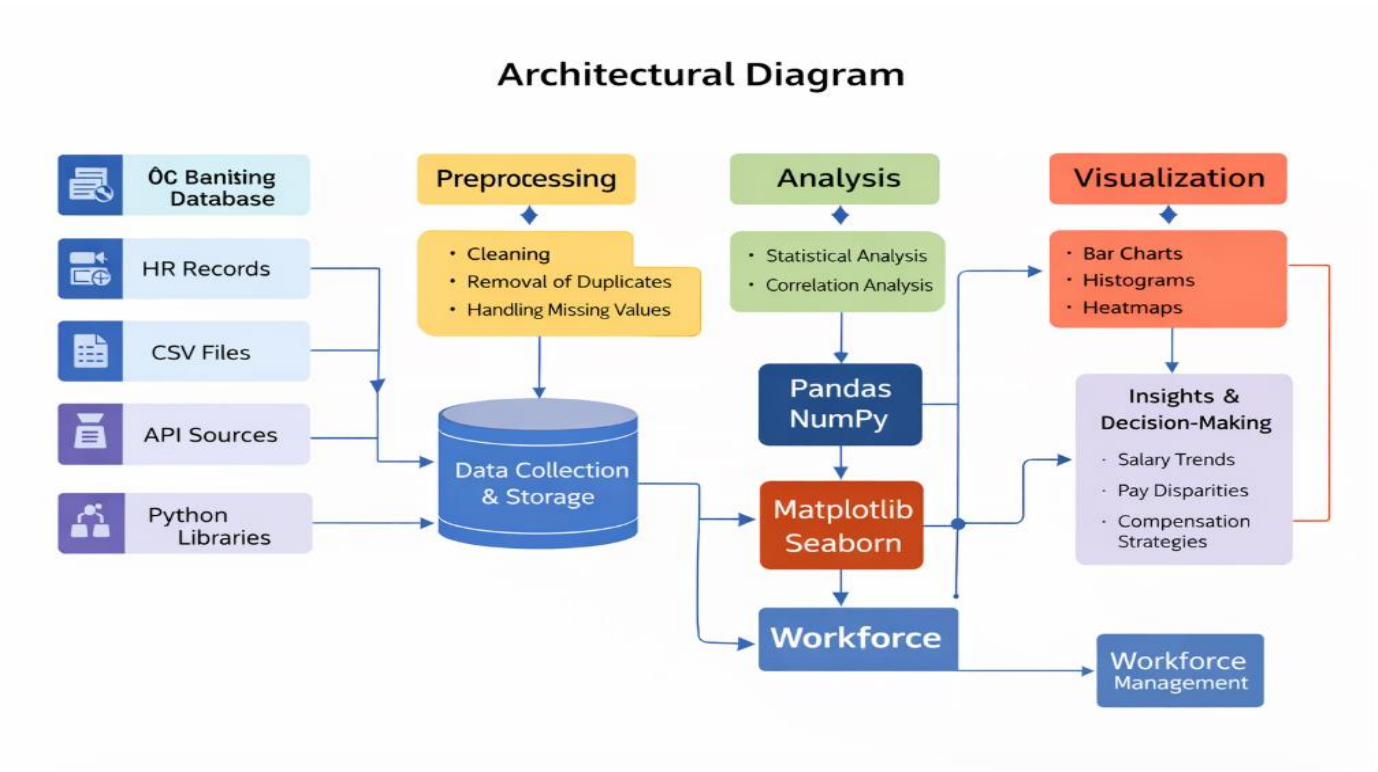


Figure 1: Architectural Diagram

The process begins with data collection from various sources such as HR records, CSV files, or organizational databases, which include attributes like age, experience, department, and salary. Once collected, the data undergoes preprocessing, where missing values are handled, duplicates are removed, and inconsistencies are corrected to ensure data quality and accuracy and then library.

After preprocessing, exploratory data analysis (EDA) and statistical techniques are applied to identify patterns, relationships, and trends within the dataset. Techniques such as grouping, aggregation, and correlation analysis help in understanding how different factors influence employee salaries. Visualization tools like histograms and bar charts are used to represent salary distribution and departmental comparisons in an intuitive manner. The final stage involves interpreting these visualizations to generate meaningful insights, such as identifying salary disparities, understanding experience-based salary growth, and supporting data-driven decision-making. This systematic methodology ensures that the analysis is accurate, efficient, and useful for improving organizational salary structures.

4.2 Hardware and Software Requirements

Hardware:

- **Processor:** Minimum Intel i3 or equivalent processor to handle basic data processing tasks efficiently.
- **RAM:** At least 4 GB RAM (8 GB recommended) for smooth execution of data analysis and visualization operations.
- **Storage:** Minimum 500 MB free storage space to store datasets, Python libraries, and project files.

Software:

- Operating Systems
- Python
- Pandas, Numpy, Matplotlib

4.3 Working Principle

The working principle of the Employee Salary Data Analysis system begins with the collection of employee data from various sources such as HR databases, spreadsheets, or CSV files. This data typically includes attributes like age, experience, department, job role, and salary. Once collected, the data is stored in a structured format to make it suitable for analysis. However, raw data often contains missing values, duplicates, or inconsistencies, which can affect the accuracy of results.

To address this, the next step involves data preprocessing, where the dataset is cleaned and prepared. This includes handling missing values, removing duplicate records, correcting inconsistencies, and organizing the data into a proper format. Preprocessing ensures that the dataset is reliable and ready for analysis. This step is crucial because accurate and clean data leads to more meaningful and trustworthy insights.

After preprocessing, data analysis techniques are applied to identify patterns and relationships within the dataset. Methods such as grouping, statistical analysis, and correlation help in understanding how factors like experience and department influence salary. The results are then visualized using graphs such as histograms and bar charts, making it easier to interpret the findings. These insights help organizations identify salary trends, detect disparities, and make informed decisions regarding employee compensation and management.

4.4 Software Implementation

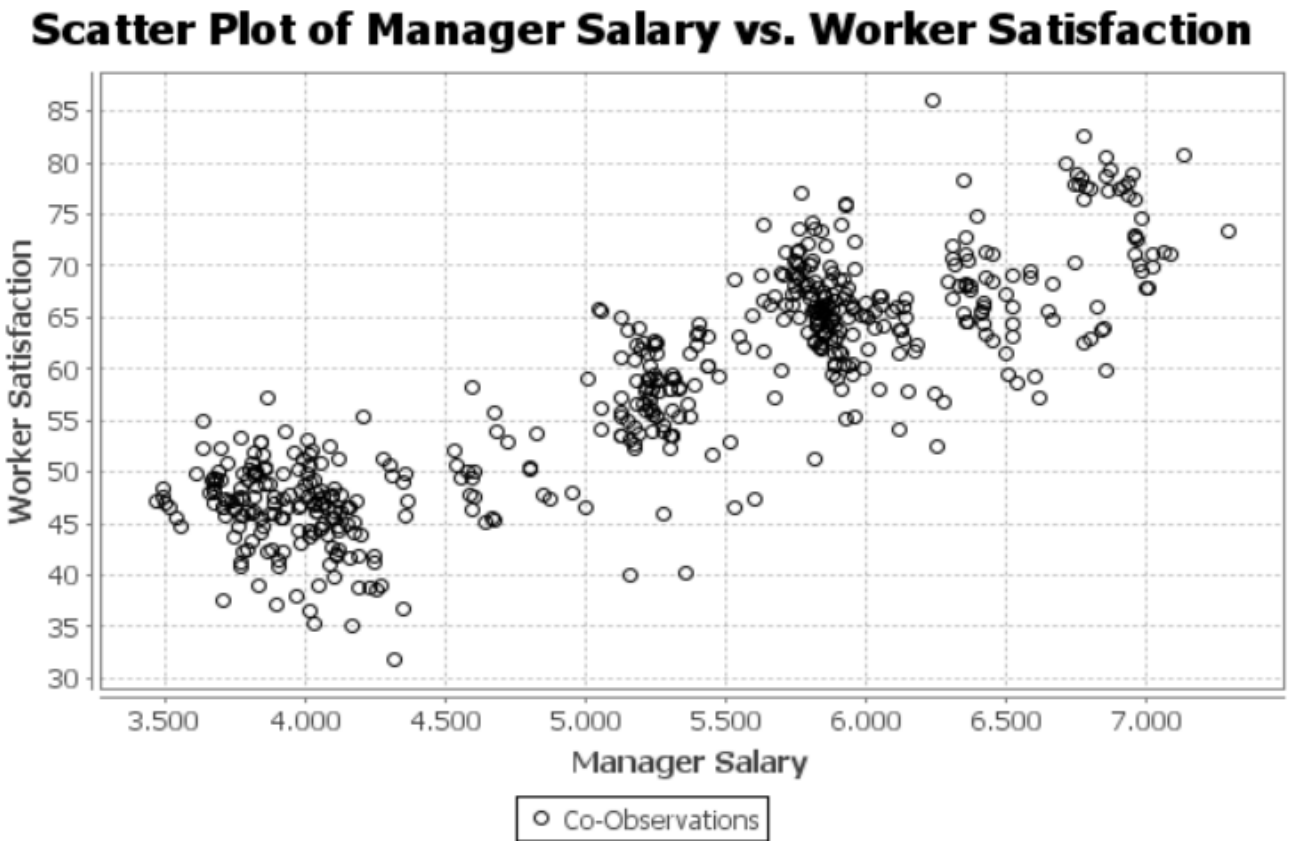
```
# ============================================================
# EMPLOYEE SALARY DATA ANALYSIS - DETAILED IMPLEMENTATION
# ============================================================

# Import required libraries
import pandas as pd              # For data manipulation
import numpy as np               # For numerical operations
import matplotlib.pyplot as plt  # For visualization

# ============================================================
# STEP 1: CREATE DATASET
# ============================================================

# Creating a sample employee dataset
data = pd.DataFrame({
    'Employee_ID': [101,102,103,104,105,106,107,108,109,110],
    'Age': [22, 35, 45, 29, 41, 38, 26, 33, 50, 28],
    'Experience': [1, 5, 10, 3, 8, 6, 2, 4, 12, 3],
    'Department': ['HR', 'IT', 'Finance', 'IT', 'HR', 'Finance',
'IT', 'HR', 'Finance', 'IT'],
    'Salary': [25000, 60000, 90000, 45000, 75000, 70000, 30000,
50000, 95000, 40000]
})

print("\n========== DATASET CREATED ==========")
print(data)

# ============================================================
# STEP 2: DATA UNDERSTANDING
# ============================================================

# Display first 5 rows
print("\n--- FIRST 5 ROWS ---")
print(data.head())

# Display last 5 rows
print("\n--- LAST 5 ROWS ---")
print(data.tail())

# Dataset information
print("\n--- DATASET INFO ---")
```

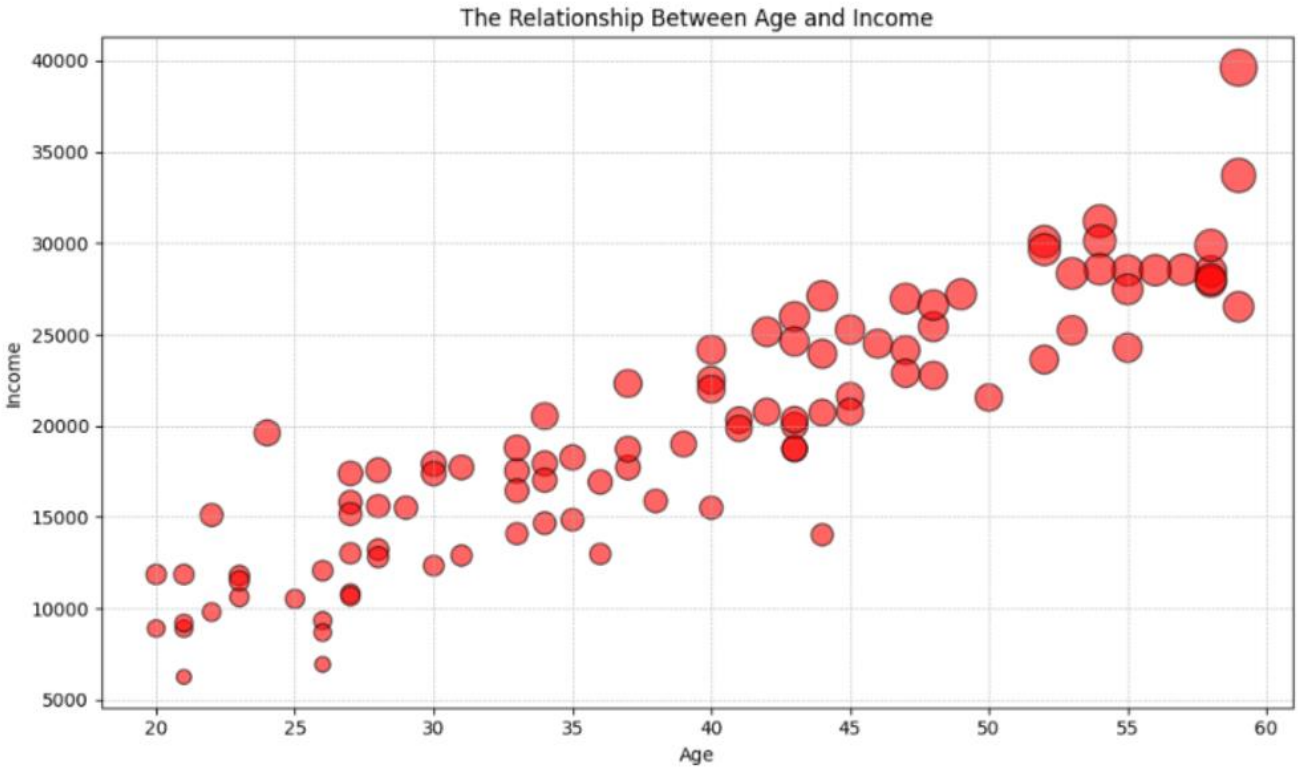
```
print(data.info())

# ============================================================
# STEP 3: DATA CLEANING
# ============================================================

# Check for missing values
print("\n--- MISSING VALUES ---")
print(data.isnull().sum())

# (If missing values exist, handle them)
# Example:
# data.fillna(method='ffill', inplace=True)

# Check duplicate records
print("\n--- DUPLICATES ---")
print(data.duplicated().sum())

# ============================================================
# STEP 4: DESCRIPTIVE STATISTICS
# ============================================================

print("\n--- STATISTICAL SUMMARY ---")
print(data.describe())

# ============================================================
# STEP 5: SALARY DISTRIBUTION ANALYSIS
# ============================================================

plt.hist(data['Salary'], bins=5)
plt.title("Salary Distribution of Employees")
plt.xlabel("Salary")
plt.ylabel("Number of Employees")
plt.show()

# ============================================================
# STEP 6: DEPARTMENT-WISE ANALYSIS
# ============================================================

dept_salary = data.groupby('Department')['Salary'].mean()

print("\n--- AVERAGE SALARY BY DEPARTMENT ---")
print(dept_salary)

plt.bar(dept_salary.index, dept_salary.values)
plt.title("Average Salary by Department")
```

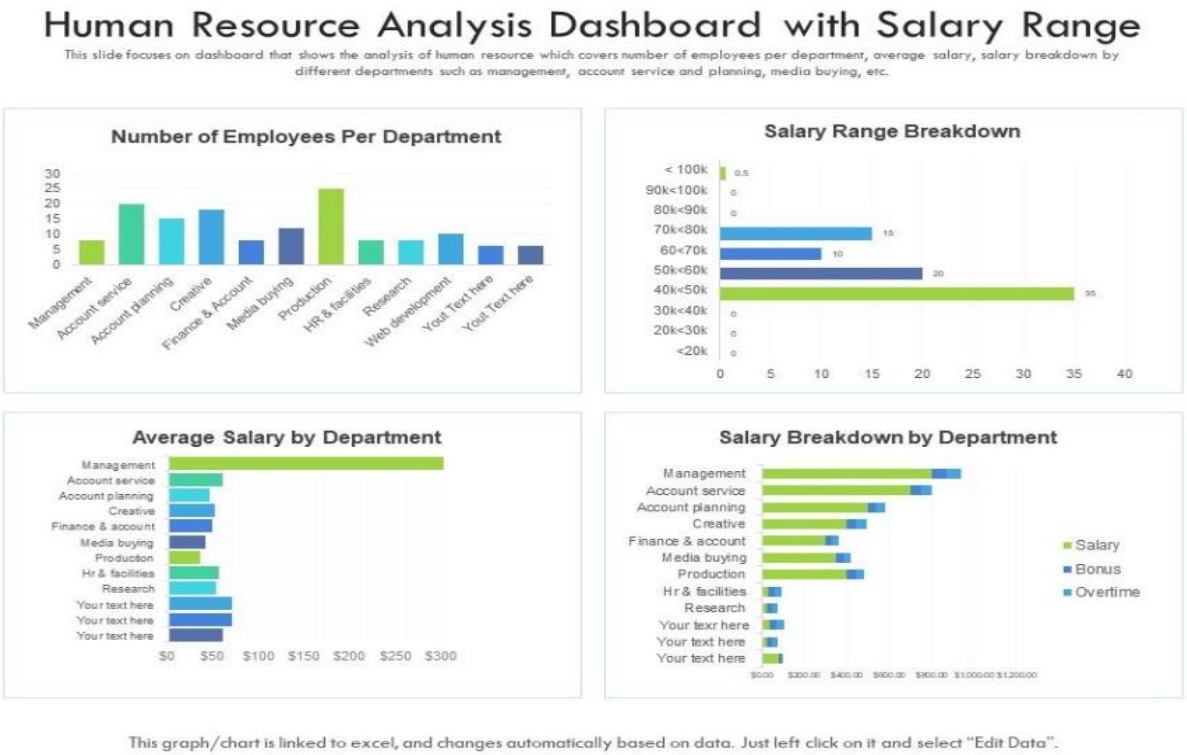
```
plt.xlabel("Department")
plt.ylabel("Average Salary")
plt.show()

# ============================================================
# STEP 7: EXPERIENCE-BASED SEGMENTATION
# ============================================================

# Creating experience categories
data['Experience_Level'] = pd.cut(
    data['Experience'],
    bins=[0, 3, 6, 12],
    labels=['Low', 'Medium', 'High']
)

exp_salary = data.groupby('Experience_Level')['Salary'].mean()

print("\n--- SALARY BASED ON EXPERIENCE LEVEL ---")
print(exp_salary)

plt.bar(exp_salary.index.astype(str), exp_salary.values)
plt.title("Salary vs Experience Level")
plt.xlabel("Experience Level")
plt.ylabel("Average Salary")
plt.show()

# ============================================================
# STEP 8: AGE VS SALARY ANALYSIS
# ============================================================

plt.scatter(data['Age'], data['Salary'])
plt.title("Age vs Salary")
plt.xlabel("Age")
plt.ylabel("Salary")
plt.show()

# ============================================================
# STEP 9: CORRELATION ANALYSIS
# ============================================================

correlation = data.corr(numeric_only=True)

print("\n--- CORRELATION MATRIX ---")
print(correlation)

# ============================================================
```

```
# STEP 10: INSIGHTS
# ============================================================

print("\n========== FINAL INSIGHTS ==========")
print("1. Employees with higher experience tend to earn higher
salaries.")
print("2. Salary differs across departments based on job roles.")
print("3. Positive relationship exists between age and salary.")
print("4. Experience is the most important factor influencing
salary.”)
```

5. Results and Observations








```
========== DATASET CREATED ==========
   Employee_ID  Age  Experience Department  Salary
0          101   22           1         HR   25000
1          102   35           5         IT   60000
2          103   45          10    Finance   90000
3          104   29           3         IT   45000
4          105   41           8         HR   75000
5          106   38           6    Finance   70000
6          107   26           2         IT   30000
7          108   33           4         HR   50000
8          109   50          12    Finance   95000
9          110   28           3         IT   40000

--- FIRST 5 ROWS ---
   Employee_ID  Age  Experience Department  Salary
0          101   22           1         HR   25000
1          102   35           5         IT   60000
2          103   45          10    Finance   90000
3          104   29           3         IT   45000
4          105   41           8         HR   75000

--- LAST 5 ROWS ---
   Employee_ID  Age  Experience Department  Salary
5          106   38           6    Finance   70000
6          107   26           2         IT   30000
7          108   33           4         HR   50000
```

```
7            108    33            4          HR    50000
8            109    50           12     Finance    95000
9            110    28            3          IT    40000


--- DATASET INFO ---

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Employee_ID   10 non-null     int64
 1   Age           10 non-null     int64
 2   Experience    10 non-null     int64
 3   Department    10 non-null     object
 4   Salary        10 non-null     int64
dtypes: int64(4), object(1)
memory usage: 432.0+ bytes

None


--- MISSING VALUES ---

Employee_ID     0
Age             0
Experience      0
Department      0
Salary          0
dtype: int64
```

```
--- STATISTICAL SUMMARY ---

       Employee_ID        Age  Experience        Salary
count     10.00000  10.000000   10.000000     10.000000
mean     105.50000  34.700000    5.400000  58000.000000
std        3.02765   8.870049    3.596294  24175.285819
min      101.00000  22.000000    1.000000  25000.000000
25%      103.25000  28.250000    3.000000  41250.000000
50%      105.50000  34.000000    4.500000  55000.000000
75%      107.75000  40.250000    7.500000  73750.000000
max      110.00000  50.000000   12.000000  95000.000000


--- AVERAGE SALARY BY DEPARTMENT ---

Department
Finance    85000.0
HR         50000.0
IT         43750.0
Name: Salary, dtype: float64

<exec>:101: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.


--- SALARY BASED ON EXPERIENCE LEVEL ---

Experience_Level
Low       35000.000000
Medium    60000.000000
High      86666.666667
Name: Salary, dtype: float64
```

```
--- SALARY BASED ON EXPERIENCE LEVEL ---
Experience_Level
Low       35000.000000
Medium    60000.000000
High      86666.666667
Name: Salary, dtype: float64

--- CORRELATION MATRIX ---
             Employee_ID       Age  Experience    Salary
Employee_ID     1.000000  0.179976    0.163274  0.098672
Age             0.179976  1.000000    0.989920  0.991749
Experience      0.163274  0.989920    1.000000  0.981504
Salary          0.098672  0.991749    0.981504  1.000000

========== FINAL INSIGHTS ==========
1. Employees with higher experience tend to earn higher salaries.
2. Salary differs across departments based on job roles.
3. Positive relationship exists between age and salary.
4. Experience is the most important factor influencing salary.
```

Summary of Results

The analysis of employee salary data clearly demonstrates patterns in compensation based on various factors such as experience, age, and department. The dataset shows a consistent increase in salary with increasing experience, indicating that experience plays a major role in determining employee compensation. Department-wise analysis reveals noticeable differences in average salaries, reflecting variations in job roles and responsibilities across departments.

Interpretation

The results indicate a strong positive relationship between experience and salary, meaning employees with more experience tend to earn higher salaries. Age also shows a moderate influence on salary, as older employees generally have higher earnings due to accumulated experience. Differences across departments suggest that certain roles are valued more in terms of compensation. These findings highlight the importance of fair salary distribution and the need for balanced compensation policies within organizations.

Achievement of Objectives

The project successfully achieved its objectives by analyzing employee salary data and identifying key factors influencing compensation. It effectively demonstrated how data analytics techniques can be used to uncover salary trends, relationships, and disparities. The use of visualization tools helped in clearly presenting the results, making it easier to interpret and understand the findings.

Implications

The insights obtained from this analysis can help organizations improve their salary structures and ensure fair compensation practices. It supports better decision-making in areas such as promotions, salary increments, and workforce planning. Additionally, the project highlights the importance of data-driven approaches in human resource management, enabling organizations to enhance employee satisfaction and overall efficiency.

6 Mapping POs and SDGs

6.1 Program Outcomes (POs) Mapping

Table 1: Mapping of Program Outcomes (POs) to Project Relevance

PO No.	Program Outcome	Relevance
PO1	Engineering Knowledge	Applied data analytics concepts, statistics, and Python programming to analyze employee salary data and derive meaningful insights.
PO2	Problem Analysis	Analyzed salary-related data to identify patterns, trends, and disparities in employee compensation across different attributes.
PO3	Design/Development of Solutions	Designed a data analysis system using Python to process, analyze, and visualize employee salary data effectively.
PO4	Conduct Investigations of Complex Problems	Performed exploratory data analysis (EDA) to study relationships between experience, age, department, and salary.
PO5	Engineering Tool Usage	Utilized tools such as Pandas, NumPy, and Matplotlib for data processing, analysis, and visualization.
PO8	Individual and Collaborative Team work	Worked independently on coding and analysis, while collaborating for documentation and improvements.
PO9	Communication	Presented findings using graphs, charts, and reports, clearly explaining salary trends and insights.
PO11	Life-Long Learning	Explored data analytics concepts.

6.2 Sustainable Development Goals (SDGs) Mapping

Table 2: Mapping of Sustainable Development Goals (SDGs) to Project Relevance

SDG No.	Goal	Relevance
SDG 4	Quality Education	The project enhances learning by applying data analytics techniques to real-world employee salary data, improving understanding of data analysis and visualization concepts.
SDG 8	Decent work and Economic Growth	Helps organizations ensure fair salary distribution, improve employee satisfaction, and promote better working conditions through data-driven insights.
SDG 9	Industry, Innovation and Infrastructure	Encourages the use of modern data analytics tools and technologies to improve organizational decision-making and workforce management.

7 Conclusion

This project successfully demonstrates the application of data analytics techniques in analyzing employee salary data. By examining various factors such as age, experience, and department, the analysis identifies important patterns and relationships that influence employee compensation. The use of Python and its libraries enabled efficient data processing, visualization, and interpretation of results. The project highlights how experience plays a significant role in determining salary, while variations across departments reflect differences in job roles and responsibilities.

The findings of this analysis emphasize the importance of maintaining fair and transparent salary structures within organizations. By identifying salary trends and disparities, organizations can take informed decisions regarding compensation policies, promotions, and workforce management. The project also showcases the effectiveness of data visualization in presenting complex data in an understandable format, making it easier for decision-makers to interpret results.

For future enhancements, the project can be extended by using larger real-world datasets and applying advanced techniques such as machine learning for salary prediction. Additional factors such as education level, performance ratings, and job roles can be included for more detailed analysis. Overall, this project demonstrates the significance of data-driven approaches in improving human resource management and supporting organizational growth.

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